

Corridor authority

which links to electrify
how much charging capacity
corridor charge per diesel truck-km
charging surcharge per kWh

infrastructure and prices

Uncertainty

freight demand
grid carbon intensity

operating response

Carrier coalition

routes and empty-container pooling
electric or diesel traction
platoon formation
responds to its own costs

Social cost the authority minimises

investment + operating cost + CO₂ at the social cost of carbon + grid stress

Q1 Does electrifying cut CO₂?

only below a grid intensity
set by the two drivetrains;
the emission-factor choice
can decide the verdict

Q2 How should the corridor be priced?

charge diesel truck-km,
discount platoons by their
energy saving, index the
charging surcharge to the grid

Q3 What should carriers share?

information to pool empty
containers: it carries most
of the collaboration gain,
platooning little